VGG16 CODE

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import the necessary libraries import tensorflow as tf
from tensorflow.keras.datasets import mnist
from tensorflow.keras.models import sequential
from tensorflow.keras.layers import Conv2D, MaxPool2D, Flatten, Dense
from tensflow.keras.optimizers import Adam
from tensorflow.keras.utils import to_categorical, plot_model
(train_x, train_y), (test_x, test_y) = mnist.load_data()
print("The size of train_x is: {}".format(train_x.shape))
print("The size of train_y is: {}".format(train_y.shape))
print("The size of test_x is: {}".format(test_x.shape))
print("The size of test_y is: {}".format(test_y.shape))
train_x = train_x.reshape(train_x.shape[0],28, 28,1)
test_x = test_x.reshape(test_x.shape[0],28,28,1)
train_x = train_x/255.0
test_x = test_x/255.0
train_y = to_categorical(train_y, num_classes=10)
test_y = to_categorical(test_y, num_classes=10)
model = sequential()
model.add(Conv2D(32, (3,3), activation='relu', padding='same',
    input_shape=(28,28,1)))
model.add(Conv2D(32,(3,3), activation='relu', padding='same'))
model.add(MaxPool2D(pool_size=(2,2), stride=2))
model.add(Conv2D(64, (3,3), activation='relu', padding='same'))
model.add(Conv2D(64,(3,3), activation='relu', padding='same'))
model.add(MaxPool2D(pool_size=(2,2), strides=2))
model.add(Conv2D(128, (3,3), activation='relu', padding=
    'same'))
model.add(MaxPool2D(pool_size=(2,2), strides=2))
model.add(Flatten())
model.add(Dense(256, activation='relu'))
model.add(Dense(10, activation='soft-max'))
model.summary().
model.compile(loss='categorical_crossentropy', optimizer=
    Adam(), metrics=['accuracy'])
test_loss_list=[]
test_accuracy_list=[]
for i in range(20):
    print("\nIteration:",i+1)
```

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model.fit(train_x, train_y, batch_size=128, epochs=1,
          verbose=1 validation_data=(test_x, test_y))
score = model.evaluate(test_x, test_y, verbose=0)
print("Test loss:", score[0])
print("Test accuracy:", score[1])
test_loss_list.apperd(score[0])
test_accuracy_list.append(score[1])
print("\nTest loss for 20 iterations:", test_loss_list)
print("Test accuracy for 20 iterations:", test_accuracy_list)
```

OUTPUT:-

```
The size of train_x is: (60000, 28, 28)
The size of train_y is: (60000,)
The size of train_x is: (10000, 28, 28)
The size of test_y is: (10000,)
model: "sequential"
Layer (type)                output shape          param#
Conv2d-2(conv2D)            (None, 28,28,32)      320
Conv2d-3(conv2D)            (None, 28,28,32)      9248
max-pooling 2d(maxpooling2D) (None, 14,14,32)     0
Conv2d-4(conv2D)            (None, 14,14,64)      18496
Conv2d-5(conv2D)            (None, 7,7,64)        36928
max-pooling2d-1(maxpooling2D) (None, 7,7,64)      0
Conv2d-6(Conv2D)            (None, 7,7,128)       73856
Conv2d-7(conv2D)            (None, 7,7,128)       147,584
max-pooling2d-2(maxpooling2D) (None, 3,3,128)     0
flatten-1(flatten)          (None, 1152)          0
dense-3(Dense)              (None, 256)           295,168
dense-4(Dense)              (None, 10)            2570
Total params: 584,170 (2.23MB)
Trainable params: 584,170 (2.23MB)
Non-trainable params: 0 (0.00B)
Iteration: 1
469/469
Test Loss: 0.032168400479172 55
Test accuracy: 0.9884999998463257
Iteration :- 2
469/469
Test loss: 0.0298898196806
```

Test accuracy : 0.99000000953674432

Iteration : 3

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Test loss : 0.0818050794303417 2

Test accuracy : 0.98944998617172 24

Iteration : 4

4691/469

Test loss : 0.0236119000532579 42

Test accuracy : 0.99190002679824 83

Iteration : 5

469/469

Test loss : 0.0238454248756170 27

Test accuracy : 0.99129998683929 44

Iteration : 6

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Test loss : 0.0267436816692352 3

Test accuracy : 0.99169999 36108821

Iteration : 7

469/469

Test loss : 0.0208318382501602 17

Test accuracy : 0.99400001764297 49

Iteration : 8

469/469

Test loss : 0.0239010592772998 81

Test accuracy : 0.99290001392364 5

Iteration : 9

4691/469

Test loss : 0.0194138512815342 7

Test accuracy : 0.99470045912 17

Iteration : 10

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Test loss : 0.0269612800329923 63

Test accuracy : 0.99269998073577 88